

Professional Summary

Multidisciplinary scientist valued for strong interpersonal skills, initiative, and organizational clarity. Adept at simplifying complex problems, improving processes, and supporting team goals through thoughtful communication and dependable follow-through.

Technical Skills

Programming & Statistical Modeling:	Developed custom Python and R Tidyverse pipelines and reproducible statistical analyses in epigenetic and proteomic research. Managed MySQL databases and implemented version control workflows using Git/GitHub for collaborative bioinformatics projects. Conducted statistical analyses including ANOVA, regression, hypothesis testing, PCA, and clustering to identify patterns in stem cell metabolism and protein-binding DNA.
Machine Learning & High-Performance Computing:	Engineered supervised ML models using Scikit-learn with KNN imputation, z-score normalization, and cross-validation for surgical decision support systems. Implemented neural network architectures with TensorFlow for pattern recognition for hip image pattern recognition. Applied unsupervised learning techniques to epigenetic datasets to enhance biological age prediction models. Executed large-scale genomic analyses on Linux Condor HTC systems using SSH remote access, managing parallel jobs for polygenic risk score calculations.
Bioinformatics Tools, Visualization & Reporting:	Designed custom Unix-based tools for genome assembly and aptamer discovery. Automated RNA-seq and NGS analysis pipelines using HISAT2 for sequence alignment with reproducible workflows. Leveraged Biopython for sequence manipulation and NCBI/UniProt databases for functional annotation. Created publication-quality and interactive visualizations using Matplotlib, Seaborn, and ggplot2 for scientific publications, presentations, and EDA. Generated R Markdown and Jupyter Notebook reports, ensuring reproducibility and understanding across research projects. Trained senior staff on custom visualization pipelines for microarray and sequencing data with thorough written guides.
Web Development & Databases:	Built full-stack PWAs using Next.js 15 with Supabase (PostgreSQL/PostGIS) for geospatial applications, implementing real-time push notifications and location-based matching algorithms. Developed production systems with Astro, React, and Tailwind CSS, integrating UTC-safe scheduling and QR-code workflows. Designed relational database schemas and optimized queries for high-traffic applications.

Education

Biomedical Data Science, MS <i>University of Wisconsin – Madison</i>	<i>Aug 2025</i> GPA: 3.7
Biomedical Engineering & Neurobiology, BS <i>University of Wisconsin – Madison</i>	<i>May 2022</i> GPA: 3.6

Professional Experience

Bioinformatics Scientist	<i>Proteovista LLC.</i>	<i>Jan 2024 - present</i>
- Architected custom Unix-based bioinformatics tools to aid aptamer discovery and genome building product design - Automated data visualization pipelines using Python and Excel VBA, processing protein-binding DNA arrays, RNA-seq, and NGS datasets, reducing processing time by 90% - Utilized bioinformatics tools (BLAST, SRA Toolkit, HISAT2) for sequence alignment and variant calling - Developed algorithms for motif matching, DNA hybridization temperatures, and more, reducing experimental burden and supporting novel breast cancer aptamer discovery - Leveraged Biopython for sequence manipulation and managed data retrieval and analysis from NCBI, UniProt, and UCSC Genome Browser databases for comparative genomics and functional annotation - Automated the webscraping of IDT to assemble a database of folded oligos and applied unsupervised learning to discover potential novel GRHL2 binding motifs - Trained senior research staff on implementation of the above tools		

Intern *Reykjavik Institute of Biomedical and Neural Engineering* Jun 2024 – Dec 2024

- Engineered semi-automated ETL pipelines using Mimics API for CT image segmentation and feature extraction, reducing processing time from 10 hours to 4 minutes.
- Implemented supervised machine learning pipeline to develop predictive models for total hip arthroplasty for surgical decision support
- Established reproducible ML workflows, ensuring model interpretability and clinical applicability
- Applied statistical validation techniques to assess model performance and evaluate surgeons' preference bias

Graduate Student Research Assistant *Qiongsi Lu Laboratory, UW – Madison* Sep 2023 – Mar 2024

- Applied unsupervised learning techniques to epigenetic datasets to enhance the predictive power of biological age estimation models
- Developed Python web scraping tool to extract and harmonize covariate data from Gene Expression Omnibus
- Executed polygenic risk score (PUMAS PRS) calculations and managed parallel computational jobs for genome-wide association studies

Research Specialist *Fujifilm Cellular Dynamics Inc* Jun 2022 – Feb 2023

- Applied stem cell culture and laboratory techniques to develop robust cell culture processes and assays for iPSC transfection, genetic engineering, and characterization
- Performed cellular and molecular characterization of engineered cell lines using flow cytometry, DNASTAR, confocal microscopy, ICC, and RT-qPCR
- Authored comprehensive technical documentation including protocols, SOPs, technical reports, and research documentation for regulatory compliance and knowledge transfer
- Collaborated with cross-functional teams to establish quality control metrics and ensure reproducibility of cell culture processes

Research Assistant *Wisconsin Institute for Discovery* Dec 2018 – Mar 2020

- Conducted experimental design, ICC/IHC, Flow Cytometry, RT quantitative PCR, and confocal microscopy to investigate the regional patterning mechanisms in pluripotent stem cell differentiation, characterizing various central nervous system tissues
- Collaborated with interdisciplinary research team to advance understanding of neural development and stem cell biology

Publications

Modular derivation of diverse, regionally discrete human posterior CNS neurons enables discovery of transcriptomic patterns *Science Advances*, 2022

DOI: <https://doi.org/10.1126/sciadv.abn7430>

7615 Genomic Consequences of GRHL2 Overexpression in ER+ Breast Cancer Cells *Journal of the Endocrine Society*, 2024

DOI: <https://doi.org/10.1210/jendso/bvae163.1809>

Personal Projects

HuckHub *Full-Stack Web Application*

Location-based progressive web application for matching ultimate frisbee partners. Built with Next.js 15, Supabase (PostgreSQL with PostGIS extension), real-time push notifications, and geospatial matching algorithms for proximity-based user connections.

Ymir Safety System *Production PWA*

Developed a production-ready progressive web application using Astro, React, and Tailwind CSS with PostgreSQL backend. Automated boat checkout workflows, safety alert systems, and maintenance tracking with QR-code integration. Implemented UTC-safe scheduling system with real-time overdue detection and bilingual member management.

Xpath *Mobile App*

A React Native mobile app with an AWS serverless backend that enables approximate location sharing to maintain weak interpersonal connections worldwide.